

RAG-Augmented Multi-Agent LLM Framework for Explainable Software Vulnerability Detection Using CWE Knowledge Bases

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The Juliet number we can stand behind

This report adds a fair Juliet pair score. It continues report 8.

What we did

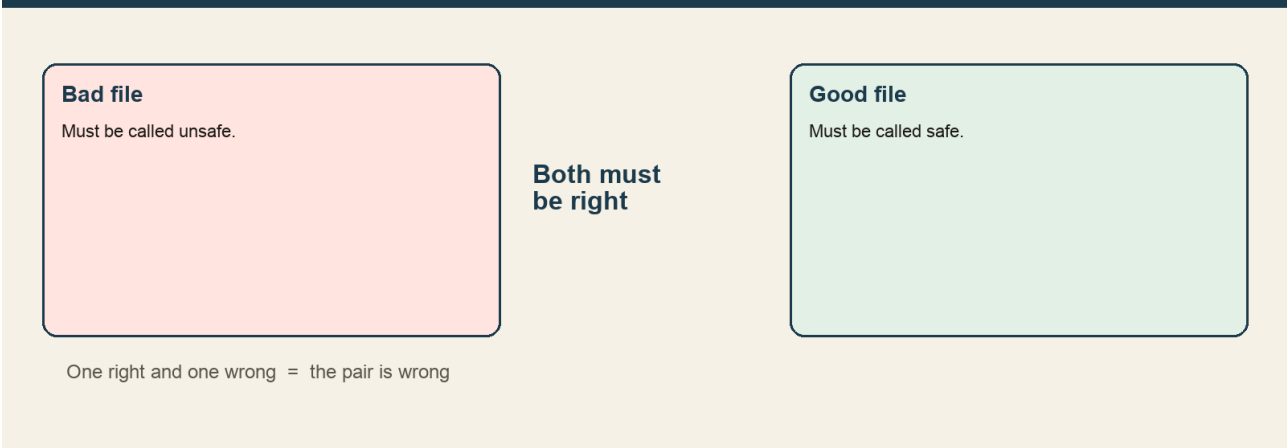
A Juliet pair is one unsafe file and one matching safe file. The model is correct only if it calls the bad file unsafe and the good file safe.

We kept 17 complete pairs. The model got 9 right. That is 0.529.

Because 17 is small, we repeated the sample many times. The score usually falls between 0.294 and 0.765. The range is wide.

One more pair was cut off by the free daily token limit. We do not count that cutoff as a model mistake. The run is partial: 35 of 36 files were scored.

Picture



Numbers for this step only

Item	Value
Complete pairs correct	9 of 17
Pair accuracy	0.529
Likely range if repeated	0.294 to 0.765
Files scored	35 of 36 (stopped by token limit)
Binary F1 on scored files	0.789

What is next

Next we will add a simple screen to run these tests, and list what is still open.

What this report does not claim

We have not yet run Juliet with CWE search turned off. We do not say that search helps on Juliet. This is not a claim of the best published score.